

### EXAMPLE 6.2

A school district decided that the number of students attending their high school was nearly unmanageable, so it was split into two districts, with District 1 students going to the old high school and District 2 students going to a newly constructed building. A group of parents became concerned with how the two districts were constructed relative to income levels. A study was thus conducted to determine whether persons in suburban District 1 have a different mean income from those in District 2. A random sample of 20 homeowners was taken in District 1. Although 20 homeowners were to be interviewed in District 2 also, one person refused to provide the information requested, even though the researcher promised to keep the interview confidential. Thus, only 19 observations were obtained from District 2. The data, recorded in thousands of dollars, produced sample means and variances as shown in Table 6.3. Use these data to construct a 95% confidence interval for  $(\mu_1 - \mu_2)$ .

TABLE 6.3

Income data for Example  
6.2

|                 | District 1 | District 2 |
|-----------------|------------|------------|
| Sample Size     | 20         | 19         |
| Sample Mean     | 18.27      | 16.78      |
| Sample Variance | 8.74       | 6.58       |

**Solution** A preliminary analysis using histograms plotted for the two samples suggests that the two populations are mound-shaped (near normal). Also, the sample variances are very similar. The difference in the sample means is

$$\bar{y}_1 - \bar{y}_2 = 18.27 - 16.78 = 1.49$$

The estimate of the common standard deviation  $\sigma$  is

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{19(8.74) + 18(6.58)}{20 + 19 - 2}} = 2.77$$

The  $t$ -percentile for  $\alpha = \alpha/2 = .025$  and  $df = 20 + 19 - 2 = 37$  is not listed in Table 2 of the Appendix, but taking the labeled value for the nearest  $df$  less than 37 ( $df = 35$ ), we have  $t_{.025} = 2.030$ . A 95% confidence interval for the difference in mean incomes for the two districts is of the form

$$\bar{y}_1 - \bar{y}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Substituting into the formula, we obtain

$$1.49 \pm 2.030(2.77) \sqrt{\frac{1}{20} + \frac{1}{19}} \quad \text{or} \quad 1.49 \pm 1.80$$

Thus, we estimate the difference in mean incomes to lie somewhere in the interval from  $-.31$  to  $3.29$ . If we multiply these limits by \$1,000, the confidence interval for the difference in mean incomes is  $-\$310$  to  $\$3,290$ . This interval includes both positive and negative values for  $\mu_1 - \mu_2$ , so we are unable to determine whether the mean income for District 1 is larger or smaller than the mean income for District 2.